

✓ 1D Arrays

1. Find the **largest and smallest** element.
2. **Reverse** an array.
3. **Sum and average** of array elements.
4. Check if the array is **palindrome**.
5. **Remove duplicates** from an array.

✓ 2D Arrays

1. **Matrix addition and subtraction**.
2. **Transpose** of a matrix.
3. **Multiplication** of two matrices.
4. Find **row-wise and column-wise sums**.
5. **Spiral printing** of a matrix (challenging).

✓ Strings

1. **Reverse** a string without using built-in functions.
2. Count **vowels, consonants, digits, and spaces**.
3. Check if a string is a **palindrome**.
4. Convert **uppercase to lowercase** manually.
5. **Remove duplicates** from a string.

✓ Characters and ASCII

1. Find the **ASCII value** of a character.
2. Print characters from **A to Z** with ASCII values.
3. **Convert case** of a character using ASCII logic.

✓ Sorting Algorithms

1. **Bubble Sort**
2. **Selection Sort**
3. **Insertion Sort**
4. Sort in **descending order**
5. Sort **names (strings)** in alphabetical order

✓ Searching in Arrays

1. **Linear Search**
2. **Binary Search** (on sorted array)
3. Search for a **character in a string**

✓ Math Logic Problems

1. **Reverse a number**
2. **Factorial** of a number
3. **Sum of digits**
4. Check if a number is **prime**
5. Check if a number is **Armstrong** or **Perfect**
6. Print **Fibonacci series**
7. Find **LCM** and **GCD**
8. **Check for palindrome number**
9. Convert **binary** ↔ **decimal**
10. **Decimal to hexadecimal** without built-ins

✓ Switch Case Programs

1. Calculator using switch
2. Day of the week (1–7)
3. Vowel or consonant checker
4. Grade calculator based on marks
5. Month name from month number

✓ Variables, Data Types, Type Casting

1. Show difference between **implicit** and **explicit casting**
2. Add `int + float`, `char + int`, etc., using casting
3. Program to demonstrate **overflow/underflow**

✓ Challenging/Advanced Practice

1. Pascal's Triangle
2. Magic Square (3x3)
3. Print a number in **words** (e.g., 123 → One Two Three)
4. Find the **second largest** element in an array
5. **Rotate a matrix** 90 degrees clockwise

✓ Logic Building with Nested Loops

1. Multiplication table (nested loop)
2. Matrix-like input and output
3. Print diagonal elements from a 2D matrix
4. Build patterns using custom symbols or input character

PATTERN 1: Square

```

**** 1234 1 2 3 4

**** 1234 5 6 7 8

**** 1234 9 10 11 12

**** 1234 13 14 15 16

```

PATTERN 2: Right angled triangle

```

*      1      1      1      1

**     12     22     21     2 3

***    123    333    321    4 5 6

****   1234   4444   4321   7 8 9 10

```

PATTERN 3:

```

****

***

**

*

```

PATTERN 4: Inverted triangle

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**** 1111

*** 222

```

** 33

* 4

PATTERN 5:

*

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PATTERN 6: Diamond

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PATTERN 7: Hollow diamond

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PATTERN : Butterfly

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PATTERN : Solid Rhombus
